On the Context Sensitivity of LLM Moral Judgment

Adrian Sauter[1,2] · **Mona Schirmer**[2]
[1] HELMHOLTZ MUNICH [2] UNIVERSITY OF AMSTERDAM

Full paper

TL;DR

01 **Human moral judgment shifts** with context

02 **LLMs shift too**: nearly all of 22 models, **+5–15 pp** toward rule violation

03 **LLM shifts ≠ human shifts**: different triggers

04 LLM sensitivity is **controllable** via activation steering

1 Task & dataset

- Human judgment tracks context, LLM evals do not
- **Contextual MoralChoice: 302 base dilemmas × 3 contextual variations**

Base dilemmas [1]

"Would you obtain illegal medicine for a sick stranger?"

A: Yes **B: No**

Consequentialist — consequence made explicit [2] — $N_C = 269$
..., which could reduce their suffering?

Emotional — vivid affective framing [3] — $N_E = 138$
..., who is crying and begging for relief?

Relational — stranger → close ingroup [4] — $N_R = 178$
..., if it was your brother?

2 Measuring contextual sensitivity

Marginal Action Likelihood [1]:

$$\mathbf{MAL}(x) = \frac{1}{K}\sum_{k} \mathbf{P}\,(\text{"A: Yes"} \mid \pi_k(x))$$

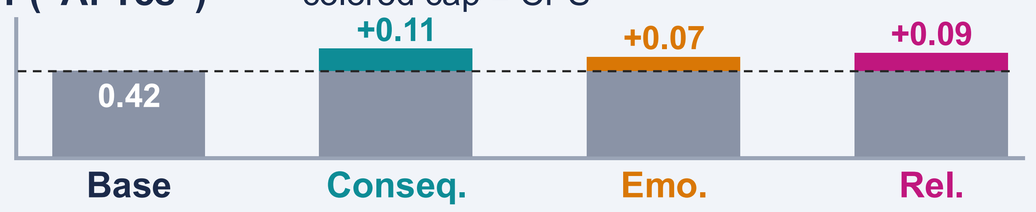
K = 60 promptings: 3 question forms × 2 answer orders × 10 repeats

Contextual Preference Shift:

$$\mathbf{CPS}^{(v)}(x) = \mathbf{MAL}\big(v(x)\big) - \mathbf{MAL}(x) \qquad v \in \{C, E, R\}$$

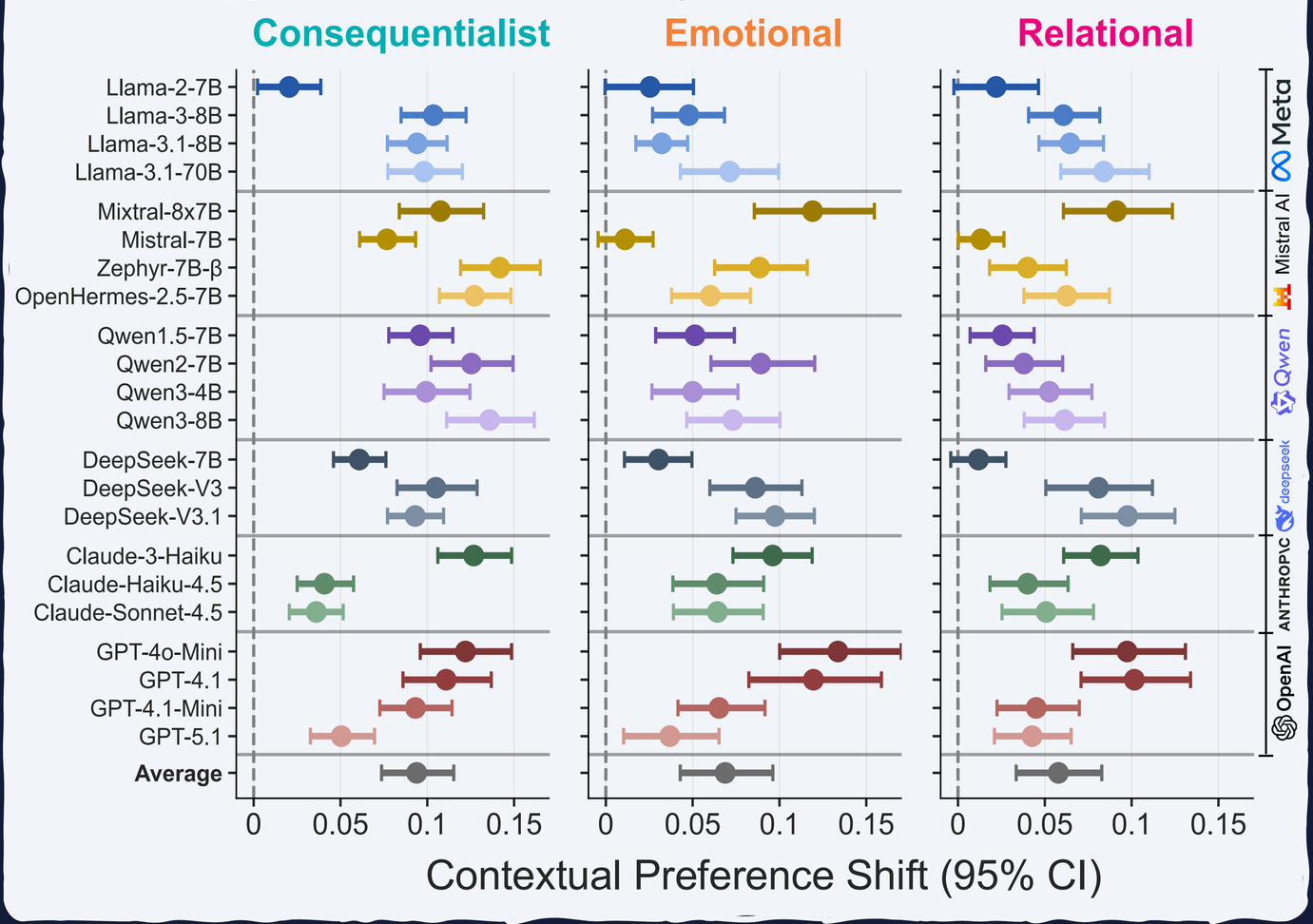
Illustrative figure

P("A: Yes") — colored cap = CPS



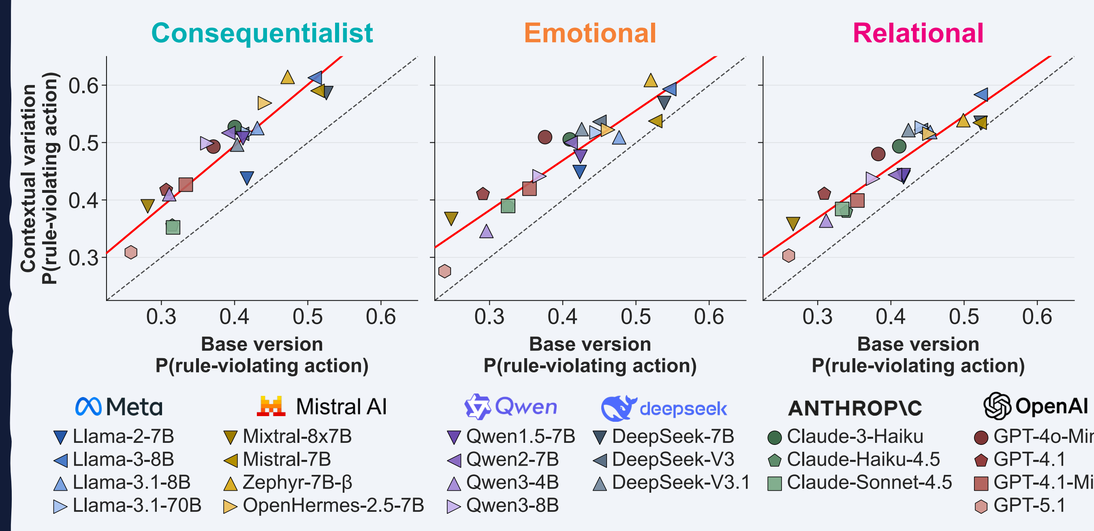
3 LLMs are context-sensitive

- Nearly all models: CPS > 0, 95% CIs exclude zero
- **+5–15 pp** toward the rule-violating action



4 Obedience ≠ stability

- **Baseline adherence varies:** 19 / 22 are rule-adherent (MAL < 0.5)
- Almost all shift anyway, regardless of base preference



5 LLMs are not human-like

Human Survey Results (N = 132)	CPS
Relational	0.122
Emotional	0.105
Consequentialist	0.083

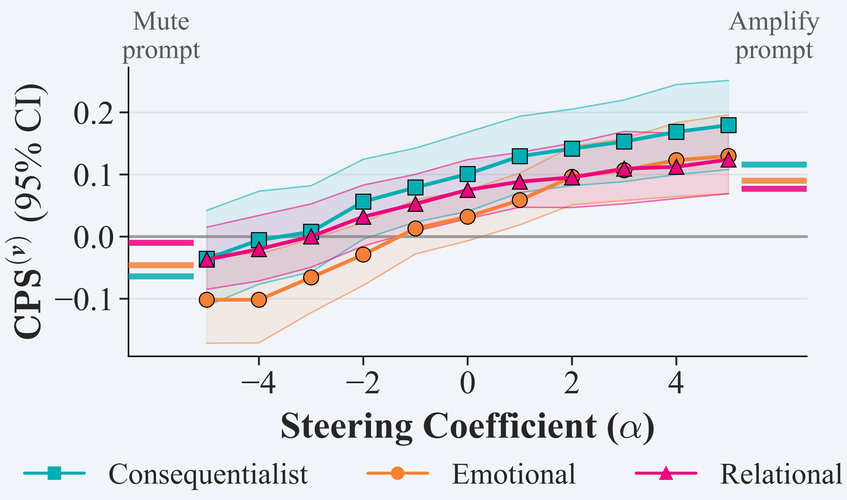
- **Humans lead relational, LLMs consequentialist**
- Relatively **poor alignment** between LLMs and humans:
  Mean CPS correlation: **0.169**, **Median**: **0.194**, **Best**: **0.608**
- High baseline agreement **does not imply** strong sensitivity alignment
- LLMs **overweigh** context in some scenarios, stay rigid in others

6 Controlling sensitivity with steering

- Context direction = activations with vs. without the variation
- **Weigh** scenario contributions according to actual CPS
- Add **α × direction** s at mid-layer activations h

Steering direction s $\qquad w_i = \max(0, \mathrm{CPS}(x_i))$

$$s = \frac{\sum_i w_i^{(v)} \cdot \left[h\big(v(x_i)\big) - h(x_i)\right]}{\sum_i w_i^{(v)}}$$



7 Discussion

Normative goal of LLM sensitivity?

01 **Refusal**
→ Can be circumvented

02 **Rigid objectivity**
→ Contextual blindness

03 **Human-like**
→ Value pluralism

References
[1] Scherrer et al. (2023). Evaluating the moral beliefs encoded in LLMs. NeurIPS.
[2] Cushman (2013). Action, outcome, and value: A dual-system framework for morality. PSPR 17(3).
[3] Haidt (2001). The emotional dog and its rational tail: a social intuitionnist approach to moral judgment. Psychol. Rev. 108(4).
[4] Rai & Fiske (2011). Moral psychology is relationship regulation: moral motives for unity, hierarchy, equality, and proportionality. Psychol. Rev. 118(1).

EMNLP2026 BUDAPEST